GraphRAG: Knowledge Graphs and Large Language Models

Graph Retrieval-Augmented Generation (GraphRAG) enhances standard LLM systems by integrating structured knowledge graphs with vector similarity retrieval.
Traditional RAG relies solely on vector databases like FAISS, Qdrant, or Pinecone, which search for chunks of text based on embedding proximity.

Neo4j is a native graph database that stores nodes and labeled relationships.
In a GraphRAG architecture, entities such as Algorithms, Concepts, and Technologies are extracted from documents using models like GPT-4o.

Backpropagation is a fundamental learning algorithm used to compute gradients for artificial neural networks during the optimization process.
Gradient Descent utilizes backpropagation to iteratively minimize the loss function.

Cosine Similarity measures the cosine of the angle between two embedding vectors, enabling fast similarity search in high-dimensional vector spaces.
The entity_embeddings vector index in Neo4j allows hybrid retrieval combining structured Cypher graph traversals and approximate nearest neighbor vector search.

FastAPI serves high-performance REST APIs connecting Streamlit frontend clients to the Neo4j database and OpenAI inference endpoints.